



Fig. 5: Salary range for validation and verification engineering professionals (Data courtesy: PayScale.com)

Expert advice

"A fresher should have undergone course work and lab exercises in digital design and VLSI design. Hands-on projects would be a big plus. Experienced engineers should have worked in the verification/validation area and gained in-depth knowledge on one or more specialisations—UVM/OVM methodologies, processor-based verification, FPGA validation, emulation and formal verification. Verification/validation has become a large collection of evolving techniques and approaches. Keeping up-to-date on the state-of-the-art techniques is important."

—**Srinivas Mandavilli, country manager, Imagination Technologies**

"Develop passion towards the field, not just look for an IT job. Money will follow you if you follow your dream. As a young boy, I used to experiment EFY circuits back when I was at school (1980's, 90's) and had been an avid follower of this wonderful magazine since then. Now I have passed on that tradition to my sons. The point I am trying to make is—follow your dreams. Enter this field only if you are passionate about electronics, and rest assured, you will cherish it for your life."

—**Srinivasan Venkataramanan, chief technology officer, CVC Pvt Ltd**

"We look for strong logical thinking and problem-solving ability and very good programming ability in potential employees. Additionally, for a fresher we look for good digital logic basics and a good understanding of computer architecture. For an experienced grad, we look for past projects, their contribution to the projects and their understanding of the projects. If you do not have programming experience, you should spend the time and effort and take some classes and learn how to program."

—**Rahul Malvi, CEO, Water Space**

"Strong electronic fundamentals, semiconductor physics and software skills such as C, C++, OOP etc are becoming key differentiators in verification job roles. For synthesis and FPGA

Usually, for beginners in this field, the pay package is up to ₹ 500,000 annually with 79 per cent jobs available for this salary range.

Juniors get anywhere between a half to one million rupees, mid-level experts draw something between one and 1.5 million and the seniors are paid anything above 1.5 million as per TimesJobs' statistics.

Talking about CVC, Venkataramanan, shares, "Our company is a VLSI ecosystem company—one that enables a vibrant VLSI ecosystem through our innovative approach to incubating role-based engineers, both freshers and the experienced alike. Hence we enable lot of freshers (approximately 150-200 engineers per year) to enter this domain.

"We also enable high-end EDA technology deployment in India (for US-based EDA start-ups) and hence we require workforce that we typically hire from freshers lot. The pay package starts around ₹ 2,40,000 per annum but big product companies pay as high as one million per annum, though they hire mostly from IITs, NITs, etc. However, it is very important to realise the steep jump in the pay package in this domain (against the likes of SW siblings). Typically, after 1-2 years experience, our own CVC employees get hired for salaries as high as ₹ 6,00,000 per annum." elnfocips too recruits freshers for this domain offering a pay package of anything between 2,00,000 and 3,00,000 rupees per annum.

Talking about Adroit IC Design, Masiwal says, "Our company is very selective in hiring freshers and we prefer candidates to have prior internship experience of at least six months. Fresh graduates' pay package varies from ₹ 4,00,000 to 6,00,000 per annum. Based on type of

roles, exposure to FPGAs helps. In terms of programming/modelling languages, HDLs such as Verilog, VHDL and HDVL such as SystemVerilog, assertion languages such as IEEE 1850 PSL, SVA are inevitable. A targeted skill for specific companies using Specman/E can be verification-specialised language such as IEEE 1647-E language. At CVC, we offer free internships for qualified students through our BUDS program."

—**Ajeetha Kumari, chief executive officer, CVC Pvt Ltd**

"As engineers, we are taught to create things. But to be a good verification engineer, you should also develop an approach to break things. Out-of-the-box thinking approach is needed. More focus should be put on learning hardware design languages such as Verilog and VHDL, digital electronics fundamentals, microprocessor and microcontroller fundamentals and programming, C++ programming skills, analytical skills and logical reasoning whilst in college."

—**Paramita Kapat, head HR, elnfocips**

"For freshers who want to join this field, my suggestion is they should focus on digital and analogue design basics, programming languages such as Verilog/C, LINUX OS and hands-on experience in a few EDA tools. These are needed because companies are very selective in hiring fresh graduates and want them to be productive as soon they join the company."

—**Deep Masiwal, CEO, Adroit IC Design**

"For becoming a verification engineer, a student should develop a solid understanding of a digital design logic works functionally. Fresh graduates are selected through our own mechanism of identifying the logical-thinking people based on the theoretical knowledge that they gain, and follow it with an interview to find out their understanding and knowledge in HDL language skills such as Verilog or System Verilog."

—**D. Srinivasan, managing director, Mobiveil**